



deti

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Evolving Piranha CMS for Contents'R'Us

**Event-Driven Extensions & Secure
Integrations**

Software Architectures, Final Project

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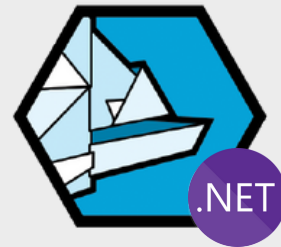


01 Overview

System Analysis

Contents'R'Us

CMS system derivated from Piranha CMS



Piranha CMS

Lightweight, cross-platform content management system built using
.NET Core

Client Demands shifted!

Asynchronous platform

Support for inbound and outbound data

Secure integration with external services

Our Scenario

Transform the CMS into a modern, **asynchronous** platform that supports inbound and outbound data flows, integrates easily with external services and leverages secure messaging



System Analysis

Disparities!!

■ Synchronous Design Bias

■ No Inbound Event Handling

Asynchronous platform

Support for inbound and outbound data

Secure integration with external services

Event driven architecture

■ Security Limitations for Integrations

■ Lack of Event Infrastructure



Project Vision

Our main vision for this assignment is to transform Piranha CMS, into a **modern, asynchronous content platform** that is integration-ready and resilient.

■ Loose Coupling ■ Secure Communication ■ Configurability

Allowing external systems to push and pull data dynamically, while maintaining:

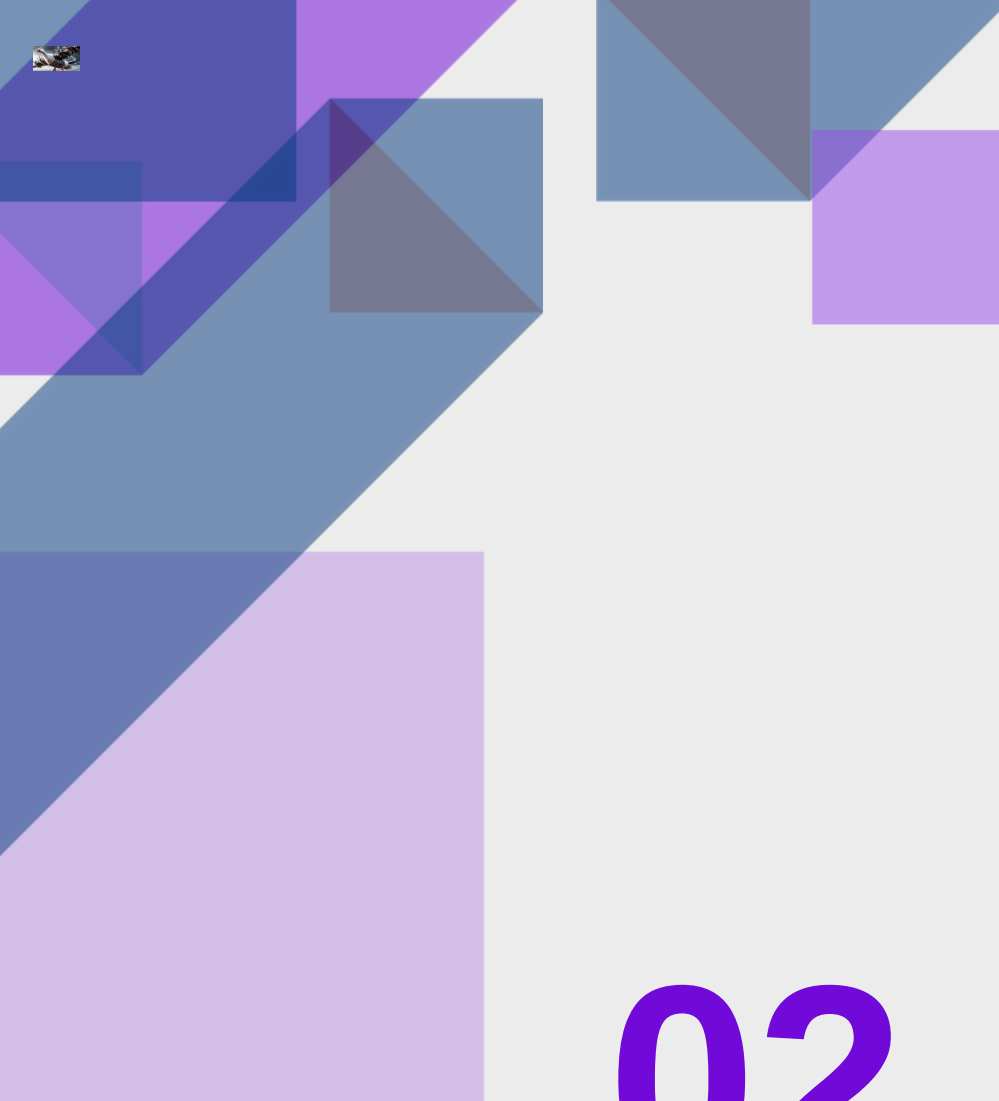
■ Trust ■ Integrity ■ Performance

Custom Publish/Subscribe Mechanism

Inbound Integration via External Publishers

Secure Configurable Messaging Infrastructure

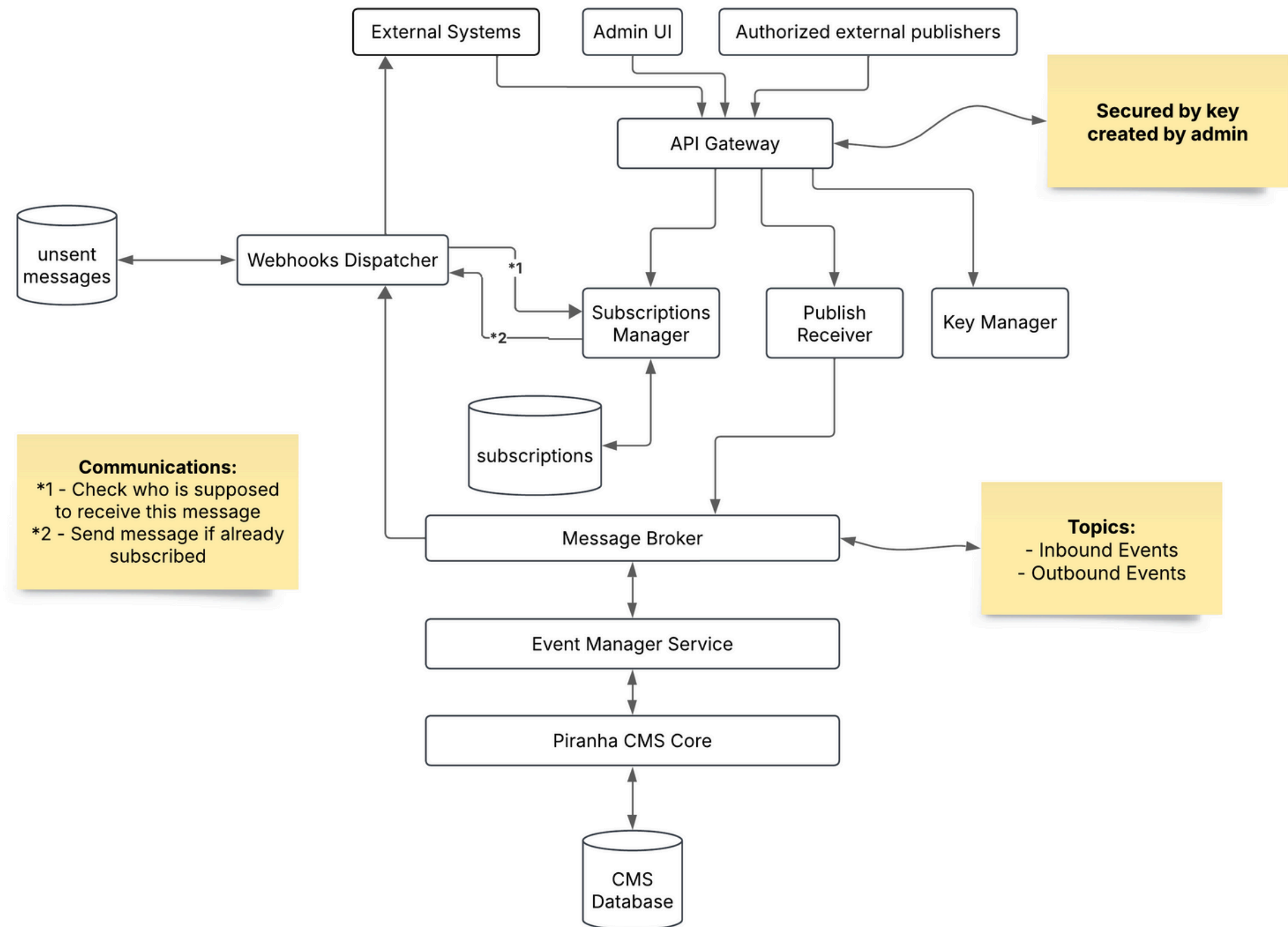




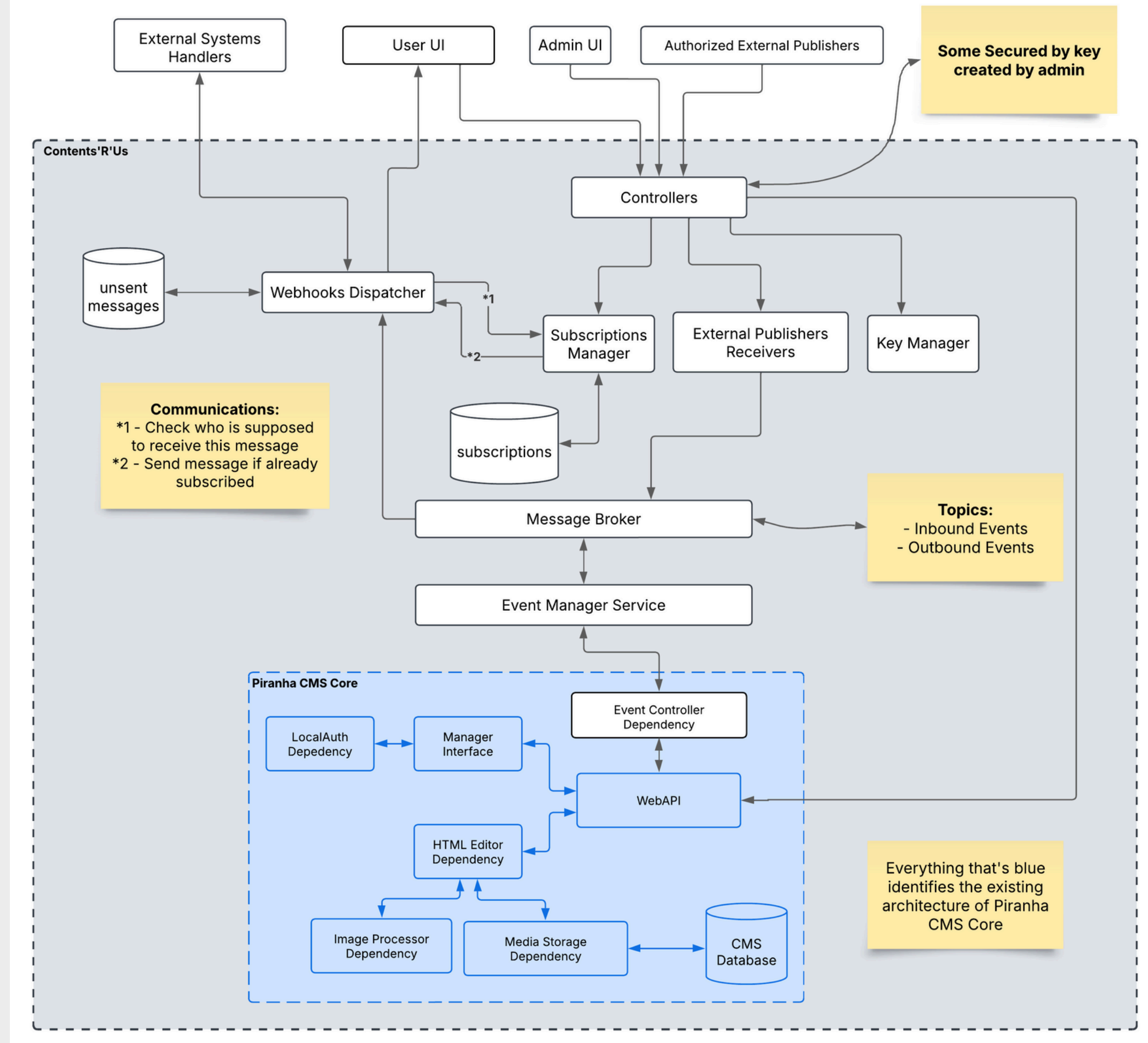
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Architecture & Proposed Roadmap

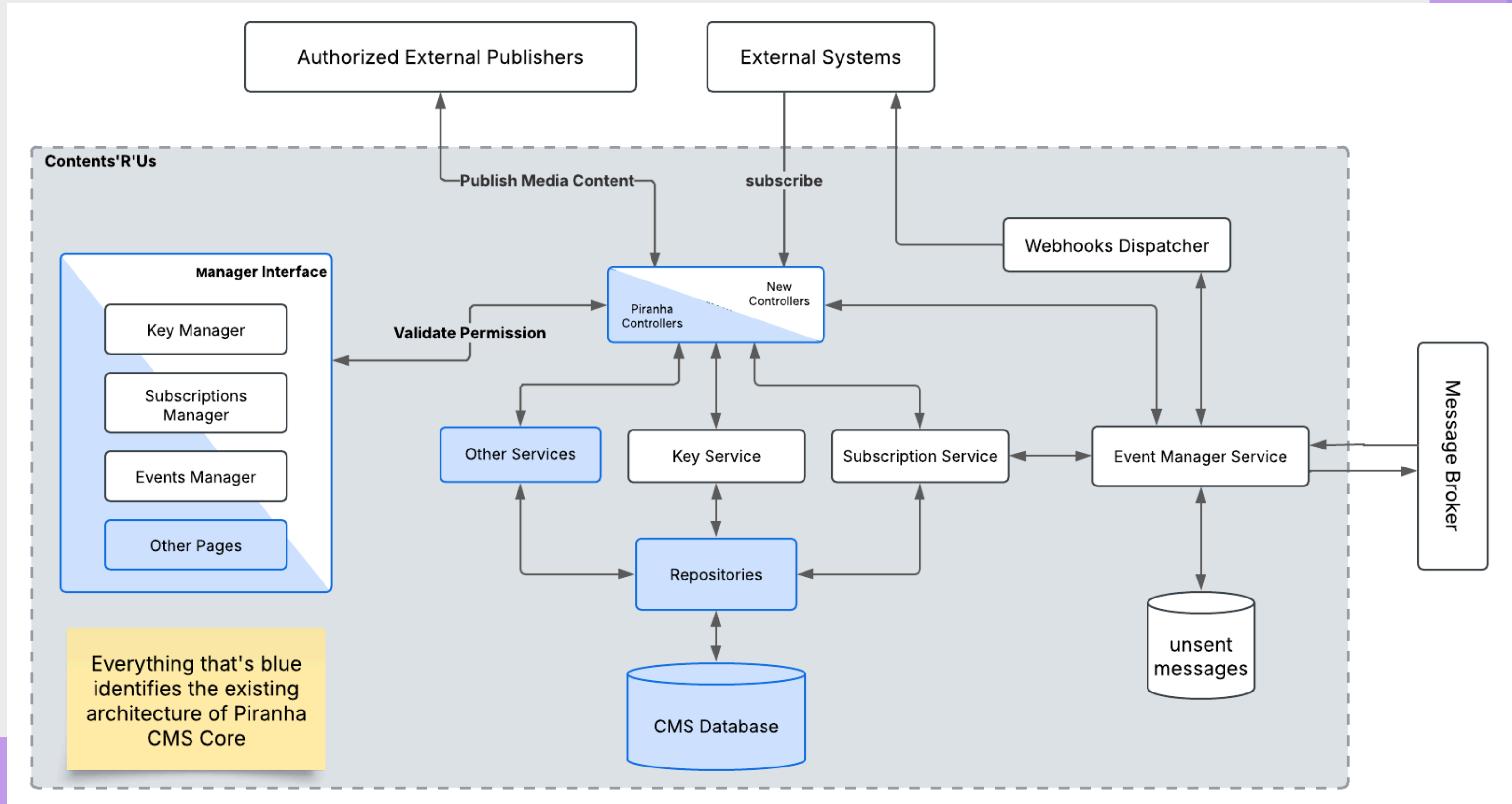
■ Envisioned Architecture



Architecture from Iteration 4



Final Architecture





■ Road Map



Event Publishing Infrastructure

Build Event Manager Service and integrate with Message Broker to capture internal events and support publishing from authorized clients.



Subscription Management Module

Implement Subscriptions Manager to handle registration, updates, deletions, and event filtering for targeted delivery.



Webhook Delivery System

Develop Webhooks Dispatcher to send event payloads to subscribers based on registered subscriptions.



EXTRA: Key Management

Create a simple key manager, to allow administrators to create keys and authorize external publishers by providing them a key



03

New features

Core Domains



Event Publishing Domain

Responsible for defining, producing, and distributing **internal domain events**.



Subscription Management Domain

Allows external systems to show interest in specific events and what type of events they wish to receive.



Security and Access Control Domain

Manages access to the publishing and integration infrastructure.



External Integration and Inbound Event Handling Domain

Enables secure, structured receipt of events from trusted external publishers.

New features



Event creation on publishing content (pages/media)



Subscriber notification on event that matches subscriber interests



Event management



Generate keys for trusted publishers add media content to the platform



Applied webhooks pattern for subscribers consume platform content



Publishers can add content through a protected endpoint



Register of subscribers trusted subscribers on the platform



Register trusted publishers



Edit subscribers interests by changing target tags



Creation of an event queue to distribute generated events



Remove publishers that are no longer considered trustworthy.



04 DEMO

Thanks

Any questions?

